Etiology and Pathogenesis

Like estrogen receptors, androgen receptors are present throughout the skin. In men, androgens are synthesized in the testes and adrenals, whereas in women, androgens are secreted by the ovaries and adrenals. In addition to testosterone, progesterone, dehydroepiandrosterone (DHEA), and DHEA sulfate (DHEA-S) are natural androgens that can lead to cutaneous manifestations when present in excess. Testosterone is converted in peripheral tissues into the more potent androgen 5α-dihydrotestosterone (5α-DHT) by the enzyme 5α-reductase. Unlike testosterone, 5α-DHT cannot be aromatized to estrogen and therefore exerts purely androgenic effects. Furthermore, 5α-DHT binds to the androgen receptor with greater affinity than testosterone.

Androgens play a key role in regulating hair growth and sebaceous gland activity. They are also involved in wound healing, epidermal differentiation, and modulation of dermal fibroblasts. The distribution of androgen receptors correlates with the development of "male-pattern" hair growth on the face, chest, and back, as well as androgenetic alopecia on the scalp. Both estrogens and androgens can bind to sex hormone-binding globulin (SHBG), also known as testosterone-estrogen-binding globulin (TEBG). SHBG levels are reduced by synthetic progestins found in oral contraceptives (e.g., norgestrel, norethindrone, norgestimate, desogestrel), as well as in conditions such as obesity, acromegaly, hypothyroidism, and hyperinsulinemia. Lower SHBG levels increase free circulating androgens, contributing to the development of androgenic symptoms in these states.

Clinical Findings and Diagnosis: Cutaneous Manifestations of Androgen Excess

Androgen-related skin disorders are common in dermatology. Although the exact pathogenesis of acne remains incompletely understood, androgen stimulation of sebaceous glands is a well-established contributing factor. Acne typically improves after adolescence in most individuals (see Chap. 78). Persistent or late-onset acne should prompt consideration of underlying androgen excess, including conditions such as congenital adrenal hyperplasia (CAH), polycystic ovary syndrome (PCOS) in women, or an androgen-secreting tumor. Rosacea and chemically induced acne should also be included in the differential diagnosis. In appropriate clinical contexts, laboratory testing can help identify the source of androgen excess (see Table 152-13). Acne is also a common manifestation of androgen excess in individuals using performance-enhancing anabolic steroids, which are testosterone derivatives.

Genetic differences influence hair follicle sensitivity to androgens, explaining the variable expression of hirsutism among women exposed to similar androgen levels.

Table 152-13 Approach to the Female Patient with Suspected Hyperandrogenism

History and physical findings suggestive of hyperandrogenism - Sudden onset of acne

- Severe acne unresponsive to standard therapy

- Irregular menstrual cycles
- Hirsutism

Laboratory evaluation (most reliable when patient is off oral contraceptives for 4–6 weeks and testing is performed before menses) - **Dehydroepiandrosterone sulfate (DHEA-S):** screens for adrenal source
- >8000 ng/mL: suggestive of adrenal tumor
- 4000–8000 ng/mL: consistent with congenital adrenal hyperplasia
- **Testosterone:** screens for ovarian source
- 150–200 ng/dL: commonly seen in polycystic ovary syndrome (PCOS)
- May also be elevated in some adrenal tumors
- **Luteinizing hormone (LH) to follicle-stimulating hormone (FSH) ratio**
- >2–3: supports diagnosis of PCOS